

Student Success Analytics Report

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Executive Summary

This project analyzed a university student dataset to identify factors associated with student success and academic performance. Microsoft Excel was used for initial data inspection; Python and DuckDB in Google Colab were used for data preparation and SQL analysis, and Tableau was used to develop interactive dashboards.

The analysis included 4,986 students and found that 1,131 were at risk of failing, while 145 were at a high risk of failing. Other findings include an average grade of 79.45, an overall pass rate of 95.72%, and an average of 10 study hours per student. Performance metrics were also analyzed across instructors, programs, and courses to identify trends and support data-driven decision-making.

This project demonstrates a complete analysis workflow, including data modeling, data preparation, SQL analysis, dashboard development, and communication through interactive visualizations.

Project Objective

The objective of this project was to analyze data regarding student success across instructors, programs, and courses at a university. The analysis wanted to answer the following questions:

- Which programs have the highest pass rate?
- What factors are most associated with student success?
- Which instructors have the highest pass rate?
- Which courses have the highest failure rates?
- Which students are at risk of failing?
- What are the retention rates by student cohort?

Dataset Overview

The dataset contains lots of tables containing information about students, instructors, courses, and programs, including:

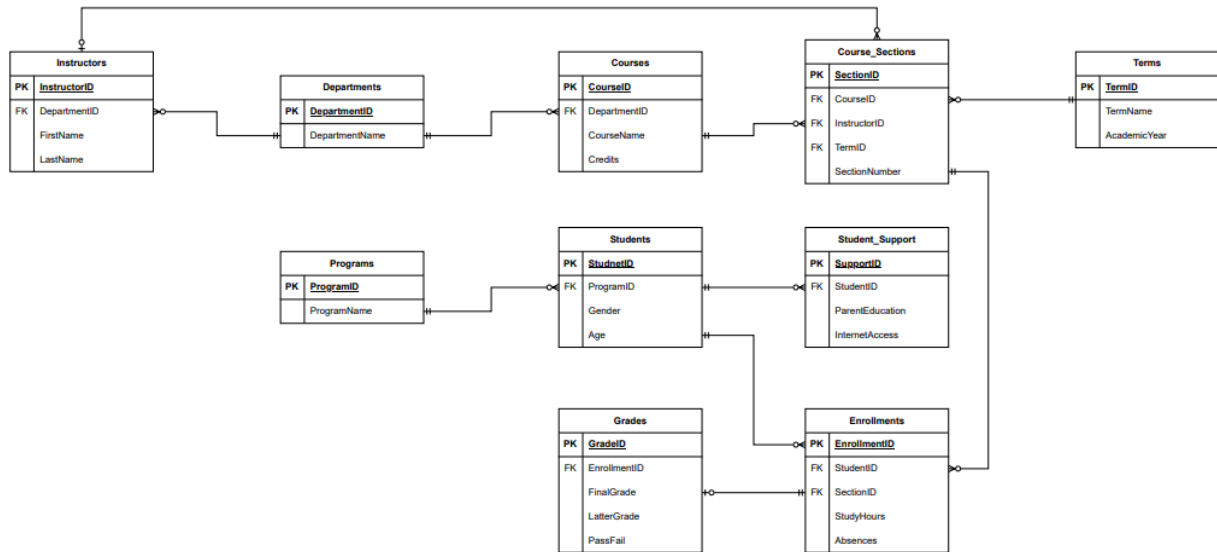
- Course Sections
- Courses offered at the university
- Departments that offer courses
- Enrollments in programs
- Grades earned by students
- Instructors who teach at the university
- Programs offered at the university
- Student Support Services
- Students
- Academic terms

The dataset was structured into multiple related tables, then analyzed using joins and aggregations in SQL.

Methodology

Data Modeling

Figure 1. Entity Relationship Diagram (ERD)



This ERD represents the relationships between all of the tables in the university dataset. This ERD was created to better understand the relationships among all the tables and to provide a blueprint for properly performing joins across multiple tables in SQL for analysis.

Data Preparation

The dataset was imported into DuckDB within Google Colab and organized into a relational database environment for analysis. During preparation, the data was reviewed to:

- Verify table relationships.
- Confirm consistency with the original CSV files.
- Ensure that joins produce accurate results.

Preparing the data in this manner ensured that the analytical queries and dashboard visualizations were based on consistent and reliable data.

SQL Analysis

While Google Colab is a Python environment, SQL queries were run through DuckDB to calculate the following metrics:

- Student Performance / Success Metrics
 - Pass Rate
 - Failure Rate
 - Retention Rate
 - At-Risk Rate
 - High-Risk Rate

- Program Metrics
 - Program Pass Rate
 - Program Failure Rate
- Instructor Metrics
 - Instructor Pass Rate
 - Instructor Failure Rate
- Course Metrics
 - Course Pass Rate
 - Course Failure Rate

The students who are considered to be “At-Risk” of failing are students whose Final grade is below 60 or below, while students who are considered to be “High-Risk” of failing are students whose final grade is below 50.

Dashboard Development

Tableau was used to create 3 interactive dashboards to visualize:

- Student Success
 - Total Students
 - At-Risk Students
 - Average Grade overall and by program
 - Pass Rate overall and by program
 - Average Study Hours
 - Study Hours vs. Final Grade
 - Enrollment by Program
 - Grade distribution
- Course and Instructor Performance
 - Total Courses
 - The total number of instructors and the number of students taught by those instructors
 - Average Pass Rate by courses and instructors
 - Average grade by instructor
- Student Risk and Performance
 - Total Students
 - At-Risk Students and by program
 - High-Risk Students and by program
 - Average Grade
 - Grade Distribution
 - Absences vs. Final Grade

Each dashboard can filter results by Fall, Spring, or Summer semesters and by academic year, ranging from 2024 to 2027.

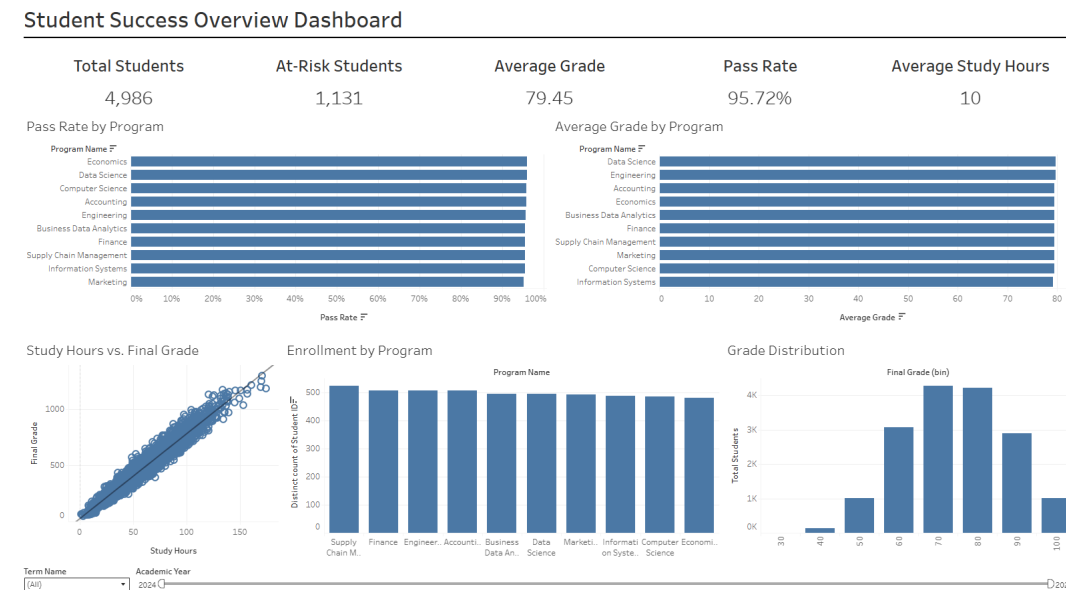
Key Performance Indicators (KPIs)

KPI	Description
Total Students	Total number of unique students included in the dataset.

Pass Rate	Percentage of students who have passed their courses.
Failure Rate	Percentage of students who have failed their courses.
Average Grade	Average final grade across all students.
Program Pass Rate	Percentage of students who have passed within their program.
Program Failure Rate	Percentage of students who have failed within their program.
Course Pass Rate	Percentage of students who have passed their courses.
Course Failure Rate	Percentage of students who have failed their courses.
Instructor Pass Rate	Percentage of students who have passed courses taught by each instructor.
Instructor Failure Rate	Percentage of students who have failed courses taught by each instructor.
Retention Rate	Percentage of students who returned for a subsequent term.
At-Risk Rate	Percentage of students whose final grade was below 60.
High-Risk Rate	Percentage of students whose final grade was below 50.

Key Findings

Figure 2. Student Success Dashboard



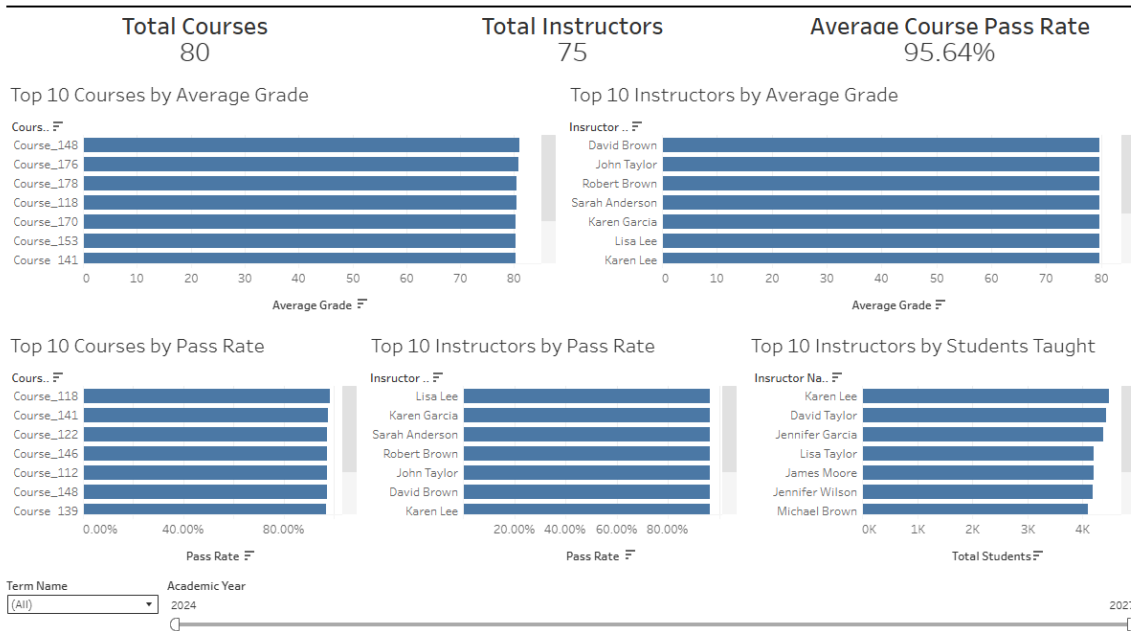
Analysis of the student success dashboard showed that the dataset contains 4,986 students, of whom 1,131 were identified as at risk of failing (final grade below 60). Students earned an average final grade of 79.45, maintained an overall pass rate of 95.72%, and studied for an average of 10 hours.

Performance across academic programs was relatively consistent. Program pass rates remained fairly consistent, ranging from 95% to 96%, and average grades remained consistent, ranging between 79.10 and 79.72. The dashboards also showed a positive relationship between study hours and final grades, indicating that students who had high study hours generally earned higher final grades. Enrollment was distributed fairly evenly across programs, and grade distribution followed an approximately bell-shaped pattern.

These findings indicate that overall academic performance is relatively consistent across programs while also identifying a group of students who may benefit from additional academic support.

Figure 3. Course and Instructor Performance Dashboard

Course and Instructor Performance Dashboard

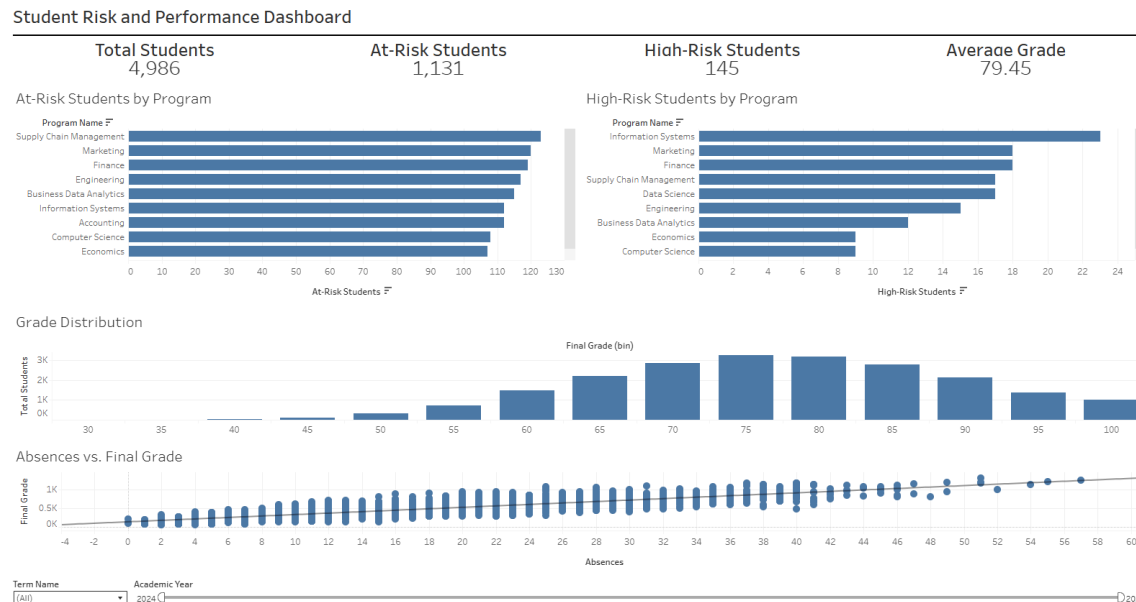


The course and instructor performance dashboard analyzed performance across 80 courses taught by 75 instructors. The overall course pass rate averaged 95.64%, while average course grades generally ranged between 80 and 81. Average grades by instructor varied only slightly, typically ranging from 79.63 to 79.73, indicating relatively consistent student performance across instructors.

Pass rates remained consistently high across both courses and instructors, although the number of students associated with each instructor varied.

These results suggest that student performance is relatively consistent throughout the university while providing an opportunity to identify courses with comparatively lower pass rates for further review.

Figure 4. Student Risk and Performance Dashboard



The student risk and performance dashboard identified 1,131 at-risk students and 145 high-risk students. One notable observation was the relationship between absences and final grades, which suggested that students with higher numbers of absences sometimes also earned higher final grades. Because this pattern differs from common expectations, additional analysis would be needed to determine whether other variables, such as study hours or course characteristics, can explain the odd relationship.

The dashboard also showed that the largest numbers of high-risk students were enrolled in the Information Systems (23 students), Marketing (18 students), and Finance (18 students) programs. These findings may help academic departments identify where additional student support or early intervention strategies could have the greatest impact.

Recommendations

1. Expand Early Intervention Programs

Identify students who are at risk early in academic terms, and provide academic support services so that students can get the help they need before they become students who have a high risk of failing. This could increase the average grade and pass rate, and even cause more and more students to come back for a subsequent semester.

2. Enhance Academic Support Services

Provide lots of ways for students to get help with their classes. Some things to have might include:

- Tutoring centers for each department, offering both in-person and online support
- Tutors for each class, offering another way to ask questions about homework and exams.
- Instructors are expanding office hours if they can make the time to have more office hours.
- AI tutors, where students can ask questions and create study plans.

- Provide free online resources for students, giving students more resources other than textbooks and lectures.

3. Share Best Practices Among Programs

Heads of each program should come together often and discuss the methods that are used for each program to help struggling students, and how instructors teach concepts that are often difficult for students.

Project Limitations

While these recommendations could help struggling students, they do have their limitations, such as:

- Students not wanting or not having the time to receive help.
- The cost of more tutoring offerings might be expensive, and offers no guarantee of helping students that much, or if students will even ask for help.
- Practices to help students in one program might not necessarily help students in another program.
- Resources that can be found online might help, but might not meet the expectations of the instructors.

Conclusion

This project provided insight into student performance across programs, courses, and instructors and identified students who may benefit from additional academic support. By combining data modeling, SQL analysis, and interactive Tableau dashboards, the project demonstrates an end-to-end business analytics workflow that transforms raw data into actionable insights.

The dashboards enable users to explore academic performance by student, instructor, course, and program by academic year and semester to easily identify trends, monitor key performance indicators, and support data-informed decision-making. Although the dataset is not from a real university, the project showcases practical analytical techniques that can be applied to real-world university data.

Appendix

Appendix A - KPI Formulas

KPI	Formula
Pass Rate	$(\text{Students Passed} \div \text{Total Students}) \times 100$
Failure Rate	$(\text{Students Failed} \div \text{Total Students}) \times 100$
Program Pass Rate	$(\text{Program Passes} \div \text{Program Enrollments}) \times 100$
Program Failure Rate	$(\text{Program Fails} \div \text{Program Enrollments}) \times 100$
Instructor Pass Rate	$(\text{Students Passed} \div \text{Students Taught}) \times 100$
Instructor Failure Rate	$(\text{Students Failed} \div \text{Students Taught}) \times 100$

Course Pass Rate	$(\text{Course Passes} \div \text{Course Enrollments}) \times 100$
Course Failure Rate	$(\text{Course Failures} \div \text{Course Enrollments}) \times 100$
Retention Rate	$(\text{Returning Students} \div \text{Original Cohort}) \times 100$
At-Risk Rate	$(\text{At-Risk Students} \div \text{Total Students}) \times 100$
High-Risk Rate	$(\text{High-Risk Students} \div \text{Total Students}) \times 100$

Appendix B - SQL Queries

Figure B1. Pass Rate

```
SELECT
    COUNT(*) AS "Total Students", -- Total Students
    SUM(CASE WHEN PassFail = 'Pass' THEN 1 ELSE 0 END) AS "Students Passed",
-- Total Students who have failed
    ROUND(SUM(CASE WHEN PassFail = 'Pass' THEN 1 ELSE 0 END) / COUNT(*) *
100.0, 2) AS "Pass Rate" -- Failure Rate = (Students Failed / Total
Students)* 100
FROM Grades
```

Figure B2. Failure Rate

```
SELECT
    COUNT(*) AS "Total Students", -- Total Students
    SUM(CASE WHEN PassFail = 'Fail' THEN 1 ELSE 0 END) AS "Studnets Failed",
-- Total Students who have failed
    ROUND(SUM(CASE WHEN PassFail = 'Fail' THEN 1 ELSE 0 END) / COUNT(*) *
100.0, 2) AS "Failure Rate" -- Failure Rate = (Studnets Failed / Total
Students)* 100
FROM Grades
```

Figure B3. Program Pass Rate

```
SELECT
    p.ProgramName, -- Program Name
    COUNT(*) AS "Total Enrollments", -- Total of studnet enrolled into
programs
    SUM(CASE WHEN g.PassFail = 'Pass' THEN 1 ELSE 0 END) AS "Program
Passes", -- Total of students who passed programs
    ROUND(SUM(CASE WHEN g.PassFail = 'Pass' THEN 1 ELSE 0 END) / COUNT(*) *
100.0, 2) AS "Program Pass Rate" -- Program Pass Rate = (Program Passes /
Program Enrollments) * 100
```

```

FROM Programs p
JOIN Students s on p.ProgramID = s.ProgramID
JOIN Enrollments e on s.StudentID = e.StudentID
JOIN Grades g on e.EnrollmentID = g.EnrollmentID
GROUP BY p.ProgramName
ORDER BY "Program Pass Rate" DESC;

```

Figure B4. Program Failure Rate

```

SELECT
    p.ProgramName, -- Program Name
    COUNT(*) AS "Total Enrollments", -- Total of studnet enrolled into
programs
    SUM(CASE WHEN g.PassFail = 'Fail' THEN 1 ELSE 0 END) AS "Program
Passes", -- Total of students who failed programs
    ROUND(SUM(CASE WHEN g.PassFail = 'Fail' THEN 1 ELSE 0 END) / COUNT(*) *
100.0, 2) AS "Program Failure Rate" -- Program Failuure Rate = (Program
Fails / Program Enrollments) * 100
FROM Programs p
JOIN Students s on p.ProgramID = s.ProgramID
JOIN Enrollments e on s.StudentID = e.StudentID
JOIN Grades g on e.EnrollmentID = g.EnrollmentID
GROUP BY p.ProgramName
ORDER BY "Program Failure Rate" DESC;

```

Figure B5. Instructor Pass Rate

```

SELECT
    CONCAT(i.FirstName, ' ', i.LastName) AS "Instructor Name", --
Instructor's Name
    ROUND(SUM(CASE WHEN g.PassFail = 'Pass' THEN 1 ELSE 0 END) / COUNT(*) *
100.0, 2) AS "Instructor Pass Rate" -- Instructor Pass Rate = (Studnets
Passed / Students Taught) * 100
FROM Instructors i
JOIN Departments d ON i.DepartmentID = d.DepartmentID
JOIN Courses c ON d.DepartmentID = c.DepartmentID
JOIN CourseSections cs ON c.CourseID = cs.CourseID
JOIN Enrollments e ON cs.SectionID = e.SectionID
JOIN Grades g ON e.EnrollmentID = g.EnrollmentID
GROUP BY i.FirstName, i.LastName

```

```
ORDER BY "Instructor Pass Rate" DESC;
```

Figure B6. Instructor Failure Rate

```
SELECT
    CONCAT(i.FirstName, ' ', i.LastName) AS "Instructor Name", --
Instructor's Name
    ROUND(SUM(CASE WHEN g.PassFail = 'Fail' THEN 1 ELSE 0 END) / COUNT(*) *
100.0, 2) AS "Instructor Failure Rate" -- Instructor Failure Rate =
(Studnets Failed / Students Taught) * 100
FROM Instructors i
JOIN Departments d ON i.DepartmentID = d.DepartmentID
JOIN Courses c ON d.DepartmentID = c.DepartmentID
JOIN CourseSections cs ON c.CourseID = cs.CourseID
JOIN Enrollments e ON cs.SectionID = e.SectionID
JOIN Grades g ON e.EnrollmentID = g.EnrollmentID
GROUP BY i.FirstName, i.LastName
ORDER BY "Instructor Failure Rate" DESC;
```

Figure B7. Course Pass Rate

```
SELECT
    c.CourseName AS "Course Name", -- Course Name
    ROUND (SUM(CASE WHEN g.PassFail = 'Pass' THEN 1 ELSE 0 END) / COUNT(*) *
100.0, 2) AS "Course Pass Rate" -- Course Pass Rate = (Passes /
Enrollemnts) * 100
FROM Courses c
JOIN CourseSections cs ON c.CourseID = cs.CourseID
JOIN Enrollments e ON cs.SectionID = e.SectionID
JOIN Grades g ON e.EnrollmentID = g.EnrollmentID
GROUP BY c.CourseName
ORDER BY "Course Pass Rate" DESC;
```

Figure B8. Course Failure Rate

```
SELECT
    c.CourseName AS "Course Name", -- Course Name
    ROUND (SUM(CASE WHEN g.PassFail = 'Fail' THEN 1 ELSE 0 END) / COUNT(*) *
100.0, 2) AS "Course Failure Rate" -- Course Failure Rate = (Fails /
Enrollemnts) * 100
FROM Courses c
```

```

JOIN CourseSections cs ON c.CourseID = cs.CourseID
JOIN Enrollments e ON cs.SectionID = e.SectionID
JOIN Grades g ON e.EnrollmentID = g.EnrollmentID
GROUP BY c.CourseName
ORDER BY "Course Failure Rate" DESC;

```

Figure B9. Retention Rate

```

-- Students who started in Fall 2024
WITH Fall2024 AS (
    SELECT DISTINCT e.StudentID
    FROM Enrollments e
    JOIN CourseSections cs ON e.SectionID = cs.SectionID
    JOIN Terms t ON cs.TermID = t.TermID
    WHERE t.TermName = 'Fall' AND t.AcademicYear = 2024
)

SELECT
    COUNT(DISTINCT f.StudentID) AS "Students in Fall 2024", -- Total
Students in Fall 2024
    COUNT(DISTINCT e.StudentID) AS "Students Returning", -- Students
Returning
    ROUND(COUNT(DISTINCT e.StudentID) / COUNT(DISTINCT f.StudentID), 2) AS
"Retention Rate" -- Retention Rate = (Students Returning / Students in
Original Cohort) * 100
FROM Fall2024 f
LEFT JOIN Enrollments e ON f.StudentID = e.StudentID

```

Figure B10. At-Risk Rate

```

SELECT
    COUNT(DISTINCT CASE
        WHEN g.FinalGrade < 60
        THEN e.StudentID
    END) AS "At-Risk Students",

    COUNT(DISTINCT e.StudentID) AS "Total Students",

    ROUND(
        100.0 *

```

```

COUNT(DISTINCT CASE
    WHEN g.FinalGrade < 60
    THEN e.StudentID
END)
/ NULLIF(COUNT(DISTINCT e.StudentID), 0),
2
) AS "At-Risk Rate"
FROM Enrollments e
JOIN Grades g
    ON e.EnrollmentID = g.EnrollmentID

```

Figure B11. High-Risk Rate

```

SELECT
    COUNT(DISTINCT CASE
        WHEN g.FinalGrade < 50
        THEN e.StudentID
    END) AS "At-Risk Students",

    COUNT(DISTINCT e.StudentID) AS "Total Students",

    ROUND(
        100.0 *
        COUNT(DISTINCT CASE
            WHEN g.FinalGrade < 50
            THEN e.StudentID
        END)
        / NULLIF(COUNT(DISTINCT e.StudentID), 0),
        2
    ) AS "At-Risk Rate"
FROM Enrollments e
JOIN Grades g
    ON e.EnrollmentID = g.EnrollmentID

```

More information on how I used SQL and DuckDB can be found in the “University_Analysis” Notebook found in the Student-Success_Analytics Repository.

Appendix C - Interactive Dashboard Links

Interactive Student Success Overview Dashboard

https://public.tableau.com/views/STUDENT_SUCCCESS_DASHBOARD/StudentSuccess?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Interactive Course and Instructor Performance Dashboard

https://public.tableau.com/shared/R8XN4BZZM?:display_count=n&:origin=viz_share_link

Interactive Student Risk and Performance Dashboard

https://public.tableau.com/views/STUDENT_RISK_AND_PERFORMANCE_DASHBOARD/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Appendix D - GitHub Repository

<https://github.com/naguthery/student-success-analytics.git>